QUAID-E-AWAM UNIVERSITY OF ENGINEERING SCIENCE & TECHNOLOGY, NAWABSHAH

BACHELOR OF SCIENCE (DATA SCIENCE)

Course Information		Course Assessment	
Title of Course:	**Discrete Mathematics**	**Theory:**	20% sessional work 20% mid-semester 60% final examination
Course Code:	MTH116		
Pre-requisite :	None		
Year / Semester:	First Year, First Semester	**Practical:**	N/A
Marks (Th + Pr):	100 + 00		
Credit Hours:	3 + 0	**Tools:**	Assignments, Tests Mid-term exams, Final Exams
Contact Hours:	45 + 00		

Course Learning Outcomes

Upon completion of this course, students will be able to achieve the following objectives:

CLOs		Domain	Taxonomy level	PLO
CLO:1	**Understand** the key concepts of Discrete Structures such as Sets, Permutations, Relations, Graphs, and Trees etc.	Cognitive	C2	1
CLO:2	**Apply** discrete structures into other computing problems such as formal specification, verification, databases, artificial intelligence, and cryptography.	Cognitive	C3	2
CLO:3	**Differentiate** various discrete structures and their relevance within the context of computer science, in the areas of data structures and algorithms, in particular.	Cognitive	C4	3

Course Contents:

Mathematical reasoning, propositional and predicate logic, rules of inference, proof by induction, proof by contraposition, proof by contradiction, proof by implication, set theory, relations, equivalence relations and partitions, partial orderings, recurrence relations, functions, mappings, function composition, inverse functions, recursive functions, Number Theory, sequences, series, counting, inclusion and exclusion principle, pigeonhole principle, permutations and combinations. Algorithms, Searching and Sorting Algorithms, elements of graph theory, planar graphs, graph colouring, Graph Algorithms, Euler graph, Hamiltonian path, rooted trees, traversals.

Textbooks:

[1] Kenneth H. Rosen, “Discrete Mathematics and Its Applications”, Latest Edition.
[2] Susanna S. Epp, “Discrete Mathematics with Applications”, Latest Edition.
[3] Richard Johnson Baugh, “Discrete Mathematics”, Latest Edition.
[4] Kolman, Busby & Ross, ”Discrete Mathematical Structures”, Latest Edition.
[5] Ralph P. Grimaldi,“Discrete and Combinatorial Mathematics: An Applied Introduction”, Latest Edition.
[6] Winifred Grassman, “Logic and Discrete Mathematics: A Computer Science Perspective”, Latest Edition.